

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
chandrusrinivasan11@cloudshell:~$ ls
README-cloudshell.txt
chandrusrinivasan11@cloudshell:~$ cd
chandrusrinivasan11@cloudshell:~$ pwd
/home/chandrusrinivasan11
chandrusrinivasan11@cloudshell:~$ mkdir folder
chandrusrinivasan11@cloudshell:~$ ls
folder README-cloudshell.txt
chandrusrinivasan11@cloudshell:~$ rmdir folder
chandrusrinivasan11@cloudshell:~$ ls
README-cloudshell.txt
chandrusrinivasan11@cloudshell:~$ touch file.txt
chandrusrinivasan11@cloudshell:~$ ls
file.txt README-cloudshell.txt
chandrusrinivasan11@cloudshell:~$ cp file.txt file1.txt
chandrusrinivasan11@cloudshell:~$ ls
file1.txt file.txt README-cloudshell.txt
chandrusrinivasan11@cloudshell:~$ mv file.txt file1.txt
chandrusrinivasan11@cloudshell:~$ ls
file1.txt README-cloudshell.txt
chandrusrinivasan11@cloudshell:~$ ln file1.txt link.txt
chandrusrinivasan11@cloudshell:~$ ls
file1.txt link.txt README-cloudshell.txt
chandrusrinivasan11@cloudshell:~$ rm file1.txt
chandrusrinivasan11@cloudshell:~$ ls
link.txt README-cloudshell.txt
chandrusrinivasan11@cloudshell:~$ rm link.txt
chandrusrinivasan11@cloudshell:~$ ls
README-cloudshell.txt
chandrusrinivasan11@cloudshell:~$ █
```

```
chandrusrinivasan11@cloudshell:~$ top
top - 10:32:07 up 22 min,  0 user,  load average: 0.02, 0.11, 0.07
Tasks:  31 total,   1 running,  30 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.5 us,   0.5 sy,   0.0 ni, 98.8 id,   0.2 wa,   0.0 hi,   0.0 si,   0.0 st
MiB Mem :   7947.4 total,   6072.7 free,   768.0 used,   1348.5 buff/cache
MiB Swap:    0.0 total,    0.0 free,    0.0 used.   7179.3 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	4324	3320	3040	S	0.0	0.0	0:00.06	bash
9	syslog	20	0	222140	3812	2456	S	0.0	0.0	0:00.01	rsyslogd
25	root	20	0	38768	27972	10868	S	0.0	0.3	0:00.50	python
26	root	20	0	6196	2620	2376	S	0.0	0.0	0:00.00	logger
68	root	10	-10	12024	4144	3032	S	0.0	0.1	0:00.00	sshd
318	root	20	0	2021836	85412	61672	S	0.0	1.0	0:00.37	dockerd
415	root	20	0	1234268	13344	8328	S	0.0	0.2	0:00.22	editor-proxy
416	root	20	0	16328	6444	5524	S	0.0	0.1	0:00.01	sudo
422	root	20	0	1226100	2700	1976	S	0.0	0.0	0:00.00	tmux-agent
449	root	20	0	1793224	45620	31852	S	0.0	0.6	0:00.59	containerd
646	root	10	-10	13996	9732	8356	S	0.0	0.1	0:00.02	sshd
651	root	20	0	2696	1576	1484	S	0.0	0.0	0:00.00	sleep
654	root	10	-10	13996	9764	8392	S	0.0	0.1	0:00.02	sshd
656	chandru+	10	-10	14256	6412	4756	S	0.0	0.1	0:00.00	sshd
657	chandru+	10	-10	7340	3740	3492	S	0.0	0.0	0:00.00	bash
660	chandru+	10	-10	7604	4460	3864	S	0.0	0.1	0:00.00	bash
663	chandru+	10	-10	14256	6448	4792	S	0.0	0.1	0:00.01	sshd
664	chandru+	10	-10	7340	3712	3464	S	0.0	0.0	0:00.00	bash
665	chandru+	10	-10	7340	3604	3344	S	0.0	0.0	0:00.00	start-shell.sh
666	chandru+	10	-10	6880	3236	2936	S	0.0	0.0	0:00.00	tmux
669	chandru+	10	-10	7148	2604	1988	S	0.0	0.0	0:00.01	tmux
670	chandru+	10	-10	8796	5812	3964	S	0.0	0.1	0:00.07	bash
712	root	20	0	9196	4008	3632	S	0.0	0.0	0:00.00	runuser

```
chandrusrinivasan11@cloudshell:~$ htop
```

```
0.0% Tasks: 31, 70 thr, 0 kthr, 1 running
0.0% Load average: 0.00 0.08 0.06
Mem 525M/1.76G Uptime: 00:24:31
Swp 0K/0K

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
1 root 20 0 432k 3320 3040 S 0.0 0.0 0:00.06 /bin/bash /google/scripts/cnrun.sh sleep infinity
2262 chandrusri 10 20 8420 5024 3916 R 0.7 0.1 0:00.39 htop: chandrusrinivasan1@pts/1.sh sleep infinity
1 root 20 0 432k 3320 3040 S 0.0 0.0 0:00.06 /bin/bash /google/scripts/cnrun.sh sleep infinity
1 root 20 0 432k 3320 3040 S 0.0 0.0 0:00.06 /bin/bash /google/scripts/cnrun.sh sleep infinity
391 root 20 0 1974M 85432 61672 S 0.0 1.0 0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
392 root 20 0 1974M 85432 61672 S 0.0 1.0 0:00.04 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
325 root 20 0 58768 22972 10868 R 0.0 0.3 0:00.53 /usr/bin/python /usr/bin/supervisord -n -c /google/devshell/supervisord.conf --pidfile=/var/run/supervisor.pid --
415 root 20 0 1205M 13344 8328 S 0.0 0.2 0:00.04 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/st
416 root 20 0 16328 6444 3524 S 0.0 0.1 0:00.01 sudo /google/devshell/tmux-agent
417 root 20 0 1974M 85496 61672 S 0.0 1.0 0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
419 root 20 0 1205M 13344 8328 S 0.0 0.2 0:00.02 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/st
420 root 20 0 1205M 13108 8328 S 0.0 0.2 0:00.05 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/st
421 root 20 0 1205M 13396 8328 S 0.0 0.2 0:00.00 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/st
420 root 20 0 1197M 2796 1976 S 0.0 0.0 0:00.06 /google/devshell/tmux-agent
420 root 20 0 1205M 13108 8328 S 0.0 0.2 0:00.06 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/st
420 root 20 0 1205M 13108 8328 S 0.0 0.2 0:00.06 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/st
421 root 20 0 1205M 13108 8328 S 0.0 0.2 0:00.00 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/st
420 root 20 0 1205M 13108 8328 S 0.0 0.2 0:00.06 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/st
421 root 20 0 1205M 13108 8328 S 0.0 0.2 0:00.00 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/st
422 root 20 0 1197M 2700 1976 S 0.0 0.0 0:00.00 /google/devshell/tmux-agentxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/st
423 root 20 0 1197M 2700 1976 S 0.0 0.0 0:00.00 /google/devshell/tmux-agent

F1:help F2:setup F3:search F4:filter F5:tree F6:sortby F7:logs F8:logs F9:kill F10:quit
```

```
chandrusrinivasan11@cloudshell:~$ free
               total        used        free      shared  buff/cache   available
Mem:           8138088       778684       6223352          1148       1384188       7359404
Swap:              0              0              0
chandrusrinivasan11@cloudshell:~$ lsb_release
No LSB modules are available.
chandrusrinivasan11@cloudshell:~$ arch
x86_64
chandrusrinivasan11@cloudshell:~$
```

```
chandrusrinivasan11@cloudshell:~$ lsb_release
No LSB modules are available.
chandrusrinivasan11@cloudshell:~$ arch
x86_64
chandrusrinivasan11@cloudshell:~$ uname
Linux
chandrusrinivasan11@cloudshell:~$ uptime
 10:39:40 up 30 min,  0 user,  load average: 0.07, 0.04, 0.04
chandrusrinivasan11@cloudshell:~$ hostname
cs-127197430132-default
chandrusrinivasan11@cloudshell:~$ dmesg
dmesg: read kernel buffer failed: Operation not permitted
chandrusrinivasan11@cloudshell:~$ free
              total          used          free      shared  buff/cache   available
Mem:           8138088       778684       6223352          1148       1384188       7359404
Swap:              0              0              0
chandrusrinivasan11@cloudshell:~$
```

Cloud Shell

Ask Gemini

shell.cloud.google.com/?show=ide%2Cterminal

Cloud Shell Editor

File Edit Selection View Go Run Terminal Help

Search

Walkthrough: Setup Code OSS Web

Get Started with Code OSS for the Web

Layout: US

0 [|||||]
1 [|||||]
Mem [|||||]
Swp [|||||]

7.3s Task: 31, 70 thr, 0 kbthr 1 running
6.1s Load average: 0.00 0.06 0.06
525M/7.76G Uptime: 00:25:14
OK/OK

PID	USER	PR	NI	VIRT	RES	SHR	B	CPUS	MEMS	TIME+	Command
453	root	20	0	1751M	45612	31652	B	0.7	0.0	0:00.13	containerd --config /var/run/docker/containerd/containerd.toml
1	root	20	0	4324	3320	3040	B	0.0	0.0	0:00.06	/bin/bash /google/scripts/onrun.sh sleep infinity
1	root	20	0	4324	3320	3040	B	0.0	0.0	0:00.06	/bin/bash /google/scripts/onrun.sh sleep infinity
391	root	20	0	1974M	85432	61672	B	0.0	1.0	0:00.00	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
392	root	20	0	1974M	85432	61672	B	0.0	1.0	0:00.04	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
325	root	20	0	30768	27972	10868	B	0.0	0.3	0:00.53	/usr/bin/python /usr/bin/supervisord -n -c /google/devshell/supervisord.conf --pidfile=/var/run/supervisor.pid --logfile=/var/
25	root	20	0	38768	27972	10868	B	0.0	0.3	0:00.54	/usr/bin/python /usr/bin/supervisord -n -c /google/devshell/supervisord.conf --pidfile=/var/run/supervisor.pid --logfile=/var/
415	root	20	0	1205M	13344	3328	B	0.0	0.2	0:00.04	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
416	root	20	0	16328	1444	3524	B	0.0	0.1	0:00.01	sudo /google/devshell/tmux-agent
417	root	20	0	1974M	85466	61672	B	0.0	1.0	0:00.00	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
419	root	20	0	1205M	13344	3328	B	0.0	0.2	0:00.02	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
420	root	20	0	1205M	13108	3328	B	0.0	0.2	0:00.05	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
421	root	20	0	1205M	13346	3328	B	0.0	0.2	0:00.00	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
420	root	20	0	1197M	2740	1976	B	0.0	0.0	0:00.06	/google/devshell/tmux-agent
420	root	20	0	1205M	13108	3328	B	0.0	0.2	0:00.06	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
420	root	20	0	1205M	13108	3328	B	0.0	0.2	0:00.05	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
421	root	20	0	1205M	13140	3328	B	0.0	0.2	0:00.00	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
421	root	20	0	1205M	13140	3328	B	0.0	0.2	0:00.06	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
421	root	20	0	1205M	13140	3328	B	0.0	0.2	0:00.00	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
422	root	20	0	1197M	2700	1976	B	0.0	0.0	0:00.00	/google/devshell/tmux-agent

F1 F2 F3 F4 F5 F6 F7 F8 F9 F10

33°C Mostly cloudy

Search

ENG IN 16:04 14-05-2026

Cloud Shell Editor

File Edit Selection View Go Run Terminal Help

Search

cloudshell x +

GNU nano 7.2 play *
nums = [1,3,2,4]

def one_bit_diff(a, b):
 x = a ^ b
 return x != 0 and (x & (x - 1)) == 0

def valid_gray(nums):
 if nums[0] != 0:
 return False

 if len(nums) != len(set(nums)):
 return False

 for i in range(len(nums) - 1):
 if not one_bit_diff(nums[i], nums[i + 1]):
 return False

 if not one_bit_diff(nums[0], nums[-1]):
 return False

 return True

print(str(valid_gray(nums)).lower())

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next Back Forward


```
chandrusrinivasan11@cloudshell:~$ mkdir msd
chandrusrinivasan11@cloudshell:~$ cd msd
chandrusrinivasan11@cloudshell:~/msd$ ls
chandrusrinivasan11@cloudshell:~/msd$ touch play
chandrusrinivasan11@cloudshell:~/msd$ nano play
chandrusrinivasan11@cloudshell:~/msd$ cat play
nums = [1,3,2,4]

def one_bit_diff(a, b):
    x = a ^ b
    return x != 0 and (x & (x - 1)) == 0

def valid_gray(nums):

    if nums[0] != 0:
        return False

    if len(nums) != len(set(nums)):
        return False

    for i in range(len(nums) - 1):
        if not one_bit_diff(nums[i], nums[i + 1]):
            return False

    if not one_bit_diff(nums[0], nums[-1]):
        return False

    return True

print(str(valid_gray(nums)).lower())
```



```
chandrusrinivasan11@cloudshell:~/msd$ more play
```

```
nums = [1,3,2,4]
```

```
def one_bit_diff(a, b):
```

```
    x = a ^ b
```

```
    return x != 0 and (x & (x - 1)) == 0
```

```
def valid_gray(nums):
```

```
    if nums[0] != 0:
```

```
        return False
```

```
    if len(nums) != len(set(nums)):
```

```
        return False
```

```
    for i in range(len(nums) - 1):
```

```
        if not one_bit_diff(nums[i], nums[i + 1]):
```

```
            return False
```

```
    if not one_bit_diff(nums[0], nums[-1]):
```

```
        return False
```

```
    return True
```

```
print(str(valid_gray(nums)).lower())
```

```
chandrusrinivasan11@cloudshell:~/msd$ tail play
```

```
    if not one_bit_diff(nums[i], nums[i + 1]):
```

```
        return False
```

```
    if not one_bit_diff(nums[0], nums[-1]):
```

```
        return False
```

```
    return True
```

```
chandrusrinivasan11@cloudshell:~/msd$ head play  
nums = [1,3,2,4]
```

```
def one_bit_diff(a, b):  
    x = a ^ b  
    return x != 0 and (x & (x - 1)) == 0
```

```
def valid_gray(nums):
```

```
    if nums[0] != 0:  
        return False
```

```
chandrusrinivasan11@cloudshell:~/msd$ ls  
play
```

```
chandrusrinivasan11@cloudshell:~/msd$ gedit  
-bash: gedit: command not found
```

```
chandrusrinivasan11@cloudshell:~/msd$ awk
Usage: mawk [Options] [Program] [file ...]
```

Program:

The -f option value is the name of a file containing program text.
If no -f option is given, a "--" ends option processing; the following parameters are the program text.

Options:

-f program-file	Program text is read from file instead of from the command-line. Multiple -f options are accepted.
-F value	sets the field separator, FS, to value.
-v var=value	assigns value to program variable var.
--	unambiguous end of options.

Implementation-specific options are prefixed with "-W". They can be abbreviated:

-W version	show version information and exit.
-W compat	pre-POSIX 2001.
-W dump	show assembler-like listing of program and exit.
-W help	show this message and exit.
-W interactive	set unbuffered output, line-buffered input.
-W exec file	use file as program as well as last option.
-W posix	stricter POSIX checking.
-W random=number	set initial random seed.
-W sprintf=number	adjust size of sprintf buffer.
-W usage	show this message and exit.

```
chandrusrinivasan11@cloudshell:~/msd$ sed
```

```
Usage: sed [OPTION]... {script-only-if-no-other-script} [input-file]...
```

```
-n, --quiet, --silent
    suppress automatic printing of pattern space
--debug
    annotate program execution
-e script, --expression=script
    add the script to the commands to be executed
-f script-file, --file=script-file
    add the contents of script-file to the commands to be executed
--follow-symlinks
    follow symlinks when processing in place
-i[SUFFIX], --in-place[=SUFFIX]
    edit files in place (makes backup if SUFFIX supplied)
-l N, --line-length=N
    specify the desired line-wrap length for the 'l' command
--posix
    disable all GNU extensions.
-E, -r, --regexp-extended
    use extended regular expressions in the script
    (for portability use POSIX -E).
-s, --separate
    consider files as separate rather than as a single,
    continuous long stream.
--sandbox
    operate in sandbox mode (disable e/r/w commands).
-u, --unbuffered
    load minimal amounts of data from the input files and flush
    the output buffers more often
-z, --null-data
    separate lines by NUL characters
--help
    display this help and exit
--version
    output version information and exit
```

```
GNU sed home page: <https://www.gnu.org/software/sed/>.
General help using GNU software: <https://www.gnu.org/gethelp/>.
chandrusrinivasan11@cloudshell:~/msd$ ls
play
chandrusrinivasan11@cloudshell:~/msd$ chmod u+x play
chandrusrinivasan11@cloudshell:~/msd$ ls
play
chandrusrinivasan11@cloudshell:~/msd$ sudo chown chandrusrinivasan play
chown: invalid user: 'chandrusrinivasan'
chandrusrinivasan11@cloudshell:~/msd$ umask
0002
chandrusrinivasan11@cloudshell:~/msd$ stat play
  File: play
  Size: 455          Blocks: 8          IO Block: 4096   regular file
Device: 8,17      Inode: 131377       Links: 1
Access: (0764/-rwxrw-r--)  Uid: ( 1000/chandrusrinivasan11)   Gid: ( 1000/chandrusrinivasan11)
Access: 2026-05-14 18:23:55.209111076 +0000
Modify: 2026-05-14 18:48:59.376753370 +0000
Change: 2026-05-14 19:04:55.449518368 +0000
 Birth: 2026-05-14 18:23:55.209111076 +0000
chandrusrinivasan11@cloudshell:~/msd$ lsattr play
-----e----- play
chandrusrinivasan11@cloudshell:~/msd$ sudo chattr +i play
chandrusrinivasan11@cloudshell:~/msd$ ls
play
chandrusrinivasan11@cloudshell:~/msd$
```

```
chandrusrinivasan11@cloudshell:~/msd$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1460
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether d6:08:42:5f:0e:c3 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1460
    inet 10.88.0.3 netmask 255.255.0.0 broadcast 10.88.255.255
    ether ea:31:df:c1:8d:ac txqueuelen 1000 (Ethernet)
    RX packets 47064 bytes 21832930 (21.8 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 39348 bytes 67852626 (67.8 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 40594 bytes 149363014 (149.3 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 40594 bytes 149363014 (149.3 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
chandrusrinivasan11@cloudshell:~/msd$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: eth0@if7: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1460 qdisc noqueue state UP group default qlen 1000
    link/ether ea:31:df:c1:8d:ac brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.88.0.3/16 brd 10.88.255.255 scope global eth0
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1460 qdisc noqueue state DOWN group default
    link/ether d6:08:42:5f:0e:c3 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
chandrusrinivasan11@cloudshell:~/msd$ ip route
default via 10.88.0.1 dev eth0
10.88.0.0/16 dev eth0 proto kernel scope link src 10.88.0.3
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
chandrusrinivasan11@cloudshell:~/msd$ ping google.com
PING google.com (74.125.68.113) 56(84) bytes of data.
64 bytes from sc-in-f113.1e100.net (74.125.68.113): icmp_seq=1 ttl=114 time=0.751 ms
64 bytes from sc-in-f113.1e100.net (74.125.68.113): icmp_seq=2 ttl=114 time=0.499 ms
64 bytes from sc-in-f113.1e100.net (74.125.68.113): icmp_seq=3 ttl=114 time=0.541 ms
64 bytes from sc-in-f113.1e100.net (74.125.68.113): icmp_seq=4 ttl=114 time=0.489 ms
64 bytes from sc-in-f113.1e100.net (74.125.68.113): icmp_seq=5 ttl=114 time=0.481 ms
64 bytes from sc-in-f113.1e100.net (74.125.68.113): icmp_seq=6 ttl=114 time=0.480 ms
64 bytes from sc-in-f113.1e100.net (74.125.68.113): icmp_seq=7 ttl=114 time=0.508 ms
64 bytes from sc-in-f113.1e100.net (74.125.68.113): icmp_seq=8 ttl=114 time=0.488 ms
```



```
64 bytes from sc-in-f113.1e100.net (74.125.68.113): icmp_seq=69 ttl=114 time=0.469 ms
64 bytes from sc-in-f113.1e100.net (74.125.68.113): icmp_seq=70 ttl=114 time=0.474 ms
64 bytes from sc-in-f113.1e100.net (74.125.68.113): icmp_seq=71 ttl=114 time=0.473 ms
64 bytes from sc-in-f113.1e100.net (74.125.68.113): icmp_seq=72 ttl=114 time=0.489 ms
^C
```

```
--- google.com ping statistics ---
```

```
72 packets transmitted, 72 received, 0% packet loss, time 72695ms
```

```
rtt min/avg/max/mdev = 0.429/0.517/0.751/0.047 ms
```

```
chandrusrinivasan11@cloudshell:~/msd$ netstat -a
```

```
Active Internet connections (servers and established)
```

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State
tcp	0	0	0.0.0.0:971	0.0.0.0:*	LISTEN
tcp	0	0	localhost:8998	0.0.0.0:*	LISTEN
tcp	0	0	0.0.0.0:ssh	0.0.0.0:*	LISTEN
tcp	0	0	localhost:33561	0.0.0.0:*	LISTEN
tcp	0	0	localhost:38051	0.0.0.0:*	LISTEN
tcp	0	0	localhost:38775	0.0.0.0:*	LISTEN
tcp	0	0	localhost:36792	localhost:ssh	TIME_WAIT
tcp	0	0	localhost:971	localhost:38492	ESTABLISHED
tcp	0	0	localhost:50314	localhost:970	ESTABLISHED
tcp	0	0	localhost:42228	localhost:ssh	ESTABLISHED
tcp	0	0	localhost:49922	localhost:ssh	TIME_WAIT

tcp	0	0	localhost:ssh	localhost:42228	ESTABLISHED
tcp	0	0	localhost:971	localhost:38500	ESTABLISHED
tcp	0	0	localhost:38500	localhost:971	ESTABLISHED
tcp	0	0	localhost:42230	localhost:ssh	ESTABLISHED
tcp	0	0	localhost:38492	localhost:971	ESTABLISHED
tcp	0	0	cs-127197430132-d:53720	sm-in-f138.1e100.:https	ESTABLISHED
tcp	0	0	localhost:ssh	localhost:42230	ESTABLISHED
tcp6	0	0	[::]:922	[::]:*	LISTEN
tcp6	0	0	[::]:900	[::]:*	LISTEN
tcp6	0	0	[::]:980	[::]:*	LISTEN
tcp6	0	0	[::]:981	[::]:*	LISTEN
tcp6	0	0	[::]:970	[::]:*	LISTEN
tcp6	0	0	[::]:ssh	[::]:*	LISTEN
tcp6	0	0	localhost:970	localhost:50314	ESTABLISHED
tcp6	0	0	cs-127197430132-def:980	10.180.213.13:49254	ESTABLISHED
tcp6	0	0	cs-127197430132-def:980	10.180.212.13:47322	ESTABLISHED
tcp6	0	0	localhost:970	localhost:50324	ESTABLISHED
tcp6	0	0	cs-127197430132-def:980	10.180.212.13:50966	ESTABLISHED
tcp6	0	0	cs-127197430132-def:980	10.180.213.13:49264	ESTABLISHED
tcp6	0	0	cs-127197430132-def:980	10.180.213.13:47372	ESTABLISHED
tcp6	0	0	cs-127197430132-def:980	10.180.212.13:47310	ESTABLISHED

Active UNIX domain sockets (servers and established)

Proto	RefCnt	Flags	Type	State	I-Node	Path
unix	3	[]	STREAM	CONNECTED	43594	
unix	3	[]	STREAM	CONNECTED	31449	
unix	2	[ACC]	STREAM	LISTENING	22261	/var/run/docker/metrics.sock
unix	3	[]	STREAM	CONNECTED	44215	
unix	2	[]	DGRAM	CONNECTED	43200	
unix	2	[]	DGRAM	CONNECTED	22872	
unix	3	[]	STREAM	CONNECTED	43597	
unix	3	[]	STREAM	CONNECTED	44220	
unix	3	[]	STREAM	CONNECTED	43233	/tmp/tmux-1000/default
unix	2	[ACC]	STREAM	LISTENING	22616	/var/run/docker/libnetwork/ecc8684cbb3c.sock
unix	3	[]	STREAM	CONNECTED	22266	
unix	3	[]	STREAM	CONNECTED	43301	
unix	3	[]	STREAM	CONNECTED	43598	
unix	3	[]	STREAM	CONNECTED	43286	
unix	2	[ACC]	STREAM	LISTENING	22926	/tmp/tmux-1000/default
unix	3	[]	STREAM	CONNECTED	43300	
unix	3	[]	STREAM	CONNECTED	43515	
unix	3	[]	STREAM	CONNECTED	42823	

```

unix 2 [ ACC ] STREAM LISTENING 36023 /tmp/vscode-ipc-d1353f5d-7dd9-4045-ae1c-8494f600e880.sock
unix 2 [ ACC ] STREAM LISTENING 36024 /tmp/vscode-ipc-79bd2e06-65b7-4f96-a7f0-2ac5c8479189.sock
unix 3 [ ] STREAM CONNECTED 43290
unix 2 [ ACC ] STREAM LISTENING 21232 /var/run/docker.sock
unix 2 [ ] STREAM CONNECTED 42807
unix 3 [ ] STREAM CONNECTED 43589
unix 2 [ ACC ] STREAM LISTENING 44205 /tmp/vscode-git-1508a68170.sock
unix 2 [ ] STREAM CONNECTED 43180
unix 3 [ ] STREAM CONNECTED 43513
unix 2 [ ] DGRAM CONNECTED 22181
unix 3 [ ] STREAM CONNECTED 43599
unix 3 [ ] STREAM CONNECTED 43222
unix 3 [ ] STREAM CONNECTED 43593
unix 3 [ ] STREAM CONNECTED 43289
unix 3 [ ] STREAM CONNECTED 43516
unix 9 [ ] DGRAM CONNECTED 21567 /dev/log
unix 3 [ ] STREAM CONNECTED 44218
unix 3 [ ] STREAM CONNECTED 43596
unix 3 [ ] STREAM CONNECTED 43221
unix 3 [ ] STREAM CONNECTED 44213
unix 3 [ ] STREAM CONNECTED 43587
unix 3 [ ] STREAM CONNECTED 43232
unix 3 [ ] STREAM CONNECTED 43584
unix 3 [ ] STREAM CONNECTED 44219
unix 3 [ ] STREAM CONNECTED 22267 /var/run/docker/containerd/containerd.sock
unix 3 [ ] STREAM CONNECTED 43514
unix 3 [ ] STREAM CONNECTED 43288
unix 3 [ ] STREAM CONNECTED 43586
unix 3 [ ] STREAM CONNECTED 42824
unix 2 [ ] DGRAM CONNECTED 21217
unix 3 [ ] STREAM CONNECTED 43512
unix 2 [ ] DGRAM CONNECTED 43216
unix 3 [ ] STREAM CONNECTED 44216
unix 3 [ ] STREAM CONNECTED 43588
unix 2 [ ] DGRAM CONNECTED 21579
unix 3 [ ] STREAM CONNECTED 41718
unix 3 [ ] STREAM CONNECTED 44217
unix 2 [ ACC ] STREAM LISTENING 22218 /var/run/docker/containerd/containerd-debug.sock
unix 2 [ ACC ] STREAM LISTENING 22219 /var/run/docker/containerd/containerd.sock.ttrpc
unix 2 [ ACC ] STREAM LISTENING 22220 /var/run/docker/containerd/containerd.sock
chandrusrinivasan11@cloudshell:~/msd$

```

```
chandrusrinivasan11@cloudshell:~/msd$ sudo apt install traceroute
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  traceroute
0 upgraded, 1 newly installed, 0 to remove and 2 not upgraded.
Need to get 60.5 kB of archives.
After this operation, 162 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 traceroute amd64 1:2.1.5-1 [60.5 kB]
Fetched 60.5 kB in 0s (176 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package traceroute.
(Reading database ... 121190 files and directories currently installed.)
Preparing to unpack .../traceroute_1%3a2.1.5-1_amd64.deb ...
Unpacking traceroute (1:2.1.5-1) ...
Setting up traceroute (1:2.1.5-1) ...
update-alternatives: using /usr/bin/traceroute.db to provide /usr/bin/traceroute (traceroute) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceroute.1.gz because associated file /usr/share/man/man1/traceroute.1.gz is not a valid gzip archive
update-alternatives: using /usr/bin/traceroute6.db to provide /usr/bin/traceroute6 (traceroute6) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceroute6.1.gz because associated file /usr/share/man/man1/traceroute6.1.gz is not a valid gzip archive
update-alternatives: using /usr/bin/lft.db to provide /usr/bin/lft (lft) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/lft.1.gz because associated file /usr/share/man/man1/lft.db.1.gz is not a valid gzip archive
update-alternatives: using /usr/bin/traceproto.db to provide /usr/bin/traceproto (traceproto) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceproto.1.gz because associated file /usr/share/man/man1/traceproto.1.gz is not a valid gzip archive
update-alternatives: using /usr/sbin/tcpttraceroute.db to provide /usr/sbin/tcpttraceroute (tcpttraceroute) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man8/tcpttraceroute.8.gz because associated file /usr/share/man/man8/tcpttraceroute.8.gz is not a valid gzip archive
Processing triggers for man-db (2.12.0-4build2) ...
chandrusrinivasan11@cloudshell:~/msd$ wget
wget: missing URL
Usage: wget [OPTION]... [URL]...

Try 'wget --help' for more options.
chandrusrinivasan11@cloudshell:~/msd$ ns lookup google.com
-bash: ns: command not found
chandrusrinivasan11@cloudshell:~/msd$
```

```
chandrusrinivasan11@cloudshell:~/msd$ dig google.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 33237
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                300     IN      A      142.250.4.139
google.com.                300     IN      A      142.250.4.102
google.com.                300     IN      A      142.250.4.138
google.com.                300     IN      A      142.250.4.101
google.com.                300     IN      A      142.250.4.100
google.com.                300     IN      A      142.250.4.113

;; Query time: 3 msec
;; SERVER: 169.254.169.254#53(169.254.169.254) (UDP)
;; WHEN: Thu May 14 19:19:51 UTC 2026
;; MSG SIZE rcvd: 135

chandrusrinivasan11@cloudshell:~/msd$
```

```
chandrusrinivasan11@cloudshell:~/msd$ ps
  PID TTY          TIME CMD
   725 pts/2    00:00:00 bash
  3818 pts/2    00:00:00 ps
chandrusrinivasan11@cloudshell:~/msd$ pgrep firefox
chandrusrinivasan11@cloudshell:~/msd$ ls
play
chandrusrinivasan11@cloudshell:~/msd$ pkill firefox
chandrusrinivasan11@cloudshell:~/msd$ ls
play
```

```
chandrusrinivasan11@cloudshell:~/msd$ ps -aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.0   4324   3428 ?        Ss   17:59   0:00 /bin/bash /google/scripts/onrun.sh sleep infinity
syslog        9  0.0  0.0  222140   3780 ?        Ssl  17:59   0:00 /usr/sbin/rsyslogd
root          25  0.0  0.3   38768  28040 ?        S    17:59   0:01 /usr/bin/python /usr/bin/supervisord -n -c /google/devshell/supervisord.conf --pidfile=/var/run/supervisor.pid --logfile=/var/log/supervisor.log
root          26  0.0  0.0    6196   2572 ?        S    17:59   0:00 logger -t supervisord
root          67  0.0  0.0   12024   4252 ?        S<   17:59   0:00 sshd: /usr/sbin/sshd -p 22 -o AuthorizedKeysFile=/etc/ssh/keys/authorized_keys [listener] 0 of 10-60 startups
root         318  0.0  1.0  2021836  84604 ?        Sl   17:59   0:01 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
root         441  0.0  0.1  1234012  12392 ?        Sl   17:59   0:01 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
root         442  0.0  0.0    16328   6352 ?        S    17:59   0:00 sudo /google/devshell/tmux-agent
root         447  0.0  0.0   1226100   2696 ?        Sl   17:59   0:00 /google/devshell/tmux-agent
root         453  0.1  0.5  1793224  46468 ?        Ssl  17:59   0:06 containerd --config /var/run/docker/containerd/containerd.toml
root          646  0.0  0.0    2696   1596 ?        S    17:59   0:00 sleep infinity
root          670  0.0  0.0    9196   4156 ?        S    17:59   0:00 runuser -l chandrusrinivasan11 -c export GOPATH=/home/chandrusrinivasan11/gopath:/google/gopath HOME=/home/chandrusrinivasan11 /bin/bash
chandru+      673  0.0  0.0    4324   3172 ?        Ss   17:59   0:00 sh ./bin/codeoss-cloudshell --host=0.0.0.0 --port=971
chandru+     695  0.2  2.0  11889832 168440 ?        Sl   17:59   0:14 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/server-main
chandru+     724  0.0  0.0    7584   3120 ?        S<   17:59   0:00 tmux new-session -A -D -n cloudshell -s 665092413
chandru+     725  0.0  0.0    8800   5788 pts/2    S<   17:59   0:00 -bash
chandru+    1599  0.2  1.0  1350424  83008 ?        Sl   18:14   0:08 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/bootstrap-f
chandru+    1610  0.0  0.0    9020   6012 pts/1    Ss+  18:14   0:00 /bin/bash --init-file /google/devshell/editor/code-oss-for-cloud-shell/out/vs/workbench/contrib/terminal/common/script
chandru+    2083  0.0  0.0    9020   6052 pts/3    Ss+  18:14   0:00 /bin/bash --init-file /google/devshell/editor/code-oss-for-cloud-shell/out/vs/workbench/contrib/terminal/common/script
chandru+    2549  0.0  0.0    9020   6112 pts/5    Ss+  18:15   0:00 /bin/bash --init-file /google/devshell/editor/code-oss-for-cloud-shell/out/vs/workbench/contrib/terminal/common/script
chandru+    2551  0.0  0.0    9020   6132 pts/6    Ss+  18:15   0:00 /bin/bash --init-file /google/devshell/editor/code-oss-for-cloud-shell/out/vs/workbench/contrib/terminal/common/script
chandru+    3452  0.0  0.7  1048516  62320 ?        Sl   18:25   0:00 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/bootstrap-f
root         3479  0.0  0.1   13996   9716 ?        S<   18:26   0:00 sshd: chandrusrinivasan11 [priv]
root         3481  0.0  0.1   13996   9612 ?        S<   18:26   0:00 sshd: chandrusrinivasan11@pts/0
chandru+    3483  0.0  0.0   14256   6452 ?        S<   18:26   0:00 sshd: chandrusrinivasan11@pts/4
chandru+    3484  0.0  0.0   14256   6372 ?        S<   18:26   0:00 sshd: chandrusrinivasan11@pts/4
chandru+    3485  0.0  0.0    7340   3796 pts/0    S<   18:26   0:00 bash -c (sudo touch /var/run/google/devshell/38775 s)
chandru+    3486  0.0  0.0    7340   3632 pts/4    S<+  18:26   0:00 bash -c if [ -f /google/devshell/start-shell.sh ]; then /google/devshell/start-shell.sh ' ' ' ' '665092413' fa
chandru+    3489  0.0  0.0    7340   3572 pts/4    S<+  18:26   0:00 /bin/bash /google/devshell/start-shell.sh 665092413 false
chandru+    3490  0.0  0.0    7604   4288 pts/0    S<+  18:26   0:00 /bin/bash --norc --noprofile
chandru+    3491  0.0  0.0    6880   3264 pts/4    S<+  18:26   0:00 tmux new-session -A -D -n cloudshell -s 665092413
chandru+    3494  1.0  9.9  44544492  806548 ?        Sl   18:26   0:34 /google/devshell/editor/code-oss-for-cloud-shell/node --dns-result-order=ipv4first /google/devshell/editor/code-oss-f
chandru+    3505  0.0  0.7  1048516  61836 ?        Sl   18:26   0:00 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/bootstrap-f
chandru+    3527  0.0  0.7  1016392  62880 ?        Sl   18:26   0:00 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/extensions/json
chandru+    3534  0.0  0.8  1943396  72532 ?        Sl   18:26   0:00 /home/chandrusrinivasan11/.cache/cloud-code/cloudcode_cli/cloudcode_cli/fba419f4/cloudcode_cli duet -trace -logtostder
chandru+    3562  0.2  2.4  32890572  203352 ?        Sl   18:26   0:09 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/extensions/gemi
chandru+    3569  0.2  7.5  1853348  616552 ?        Sl   18:26   0:09 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/extensions/clo
chandru+    3837  0.0  0.0   10888   4800 pts/2    R<+  19:22   0:00 ps -aux
```

```
chandrusrinivasan11@cloudshell:~/msd$ bg
-bash: bg: current: no such job
chandrusrinivasan11@cloudshell:~/msd$ fg
-bash: fg: current: no such job
chandrusrinivasan11@cloudshell:~/msd$ jobs
chandrusrinivasan11@cloudshell:~/msd$ ls
play
chandrusrinivasan11@cloudshell:~/msd$ nice -n 10 python play
false
chandrusrinivasan11@cloudshell:~/msd$ renice 5 -p 1234
renice: failed to get priority for 1234 (process ID): No such process
chandrusrinivasan11@cloudshell:~/msd$ █
```



```
chandrusrinivasan11@cloudshell:~/msd$ df
Filesystem                1K-blocks      Used Available Use% Mounted on
overlay                  98831908 76313928  22501596   78% /
tmpfs                    65536      0        65536    0% /dev
/dev/sda1                98831908 76313928  22501596   78% /root
/dev/disk/by-id/google-home-part1 5018272 153660  4586136    4% /home
/dev/root                2019696 1288424   731272   64% /usr/lib/modules
shm                      65536      0        65536    0% /dev/shm
chandrusrinivasan11@cloudshell:~/msd$ du -h
8.0K      .
chandrusrinivasan11@cloudshell:~/msd$ sudo mkfs.ext4
Usage: mkfs.ext4 [-c|-l filename] [-b block-size] [-C cluster-size]
        [-i bytes-per-inode] [-I inode-size] [-J journal-options]
        [-G flex-group-size] [-N number-of-inodes] [-d root-directory]
        [-m reserved-blocks-percentage] [-o creator-os]
        [-g blocks-per-group] [-L volume-label] [-M last-mounted-directory]
        [-O feature[,...]] [-r fs-revision] [-E extended-option[,...]]
        [-t fs-type] [-T usage-type ] [-U UUID] [-e errors_behavior] [-z undo_file]
        [-jncvDFSV] device [blocks-count]
chandrusrinivasan11@cloudshell:~/msd$ mount
overlay on / type overlay (rw,relatime,lowerdir=206/fs:199/fs:198/fs:196/fs:193/fs:192/fs:190/fs:189/fs:188/fs:187/fs:185/fs:184/fs:183/fs:182/fs:181/fs:180/s:174/fs:173/fs:172/fs:163/fs:162/fs:161/fs:160/fs:157/fs:156/fs:155/fs:152/fs:151/fs:148/fs:146/fs:143/fs:142/fs:141/fs:140/fs:139/fs:138/fs:137/fs:136/fs:3/fs:119/fs:118/fs:116/fs:114/fs:110/fs:107/fs:106/fs:105/fs:104/fs:102/fs:99/fs:97/fs:95/fs:94/fs:91/fs:90/fs:85/fs:84/fs:83/fs:82/fs:80/fs:78/fs:69/fs:68/s:6/fs,upperdir=/var/lib/containerd/io.containerd.snapshotter.v1.gcfs/snapshotter/snapshots/317/fs,workdir=/var/lib/containerd/io.containerd.snapshotter.v1.proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev type tmpfs (rw,nosuid,size=65536k,mode=755)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=620,ptmxmode=666)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
sysfs on /sys type sysfs (ro,nosuid,nodev,noexec,relatime)
cgroup on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime)
/dev/sda1 on /root type ext4 (rw,relatime,commit=30)
/dev/disk/by-id/google-home-part1 on /home type ext4 (rw,noatime,journal_checksum,errors=remount-ro,data=ordered)
/dev/dm-0 on /usr/lib/modules type ext2 (ro,relatime)
/dev/sda1 on /etc/hosts type ext4 (rw,relatime,commit=30)
/dev/sda1 on /etc/hostname type ext4 (rw,relatime,commit=30)
/dev/sda1 on /etc/resolv.conf type ext4 (rw,relatime,commit=30)
shm on /dev/shm type tmpfs (rw,relatime,size=65536k)
/dev/sda1 on /var/config/tmux type ext4 (ro,relatime,commit=30)
/dev/sda1 on /etc/ssh/keys type ext4 (rw,relatime,commit=30)
/dev/sda1 on /var/lib/docker type ext4 (rw,relatime,commit=30)
/dev/sda1 on /run/google/devshell type ext4 (rw,relatime,commit=30)
```

```
/dev/sda1 on /etc/ssh/ssh_known_hosts type ext4 (rw,relatime,commit=30)
shm on /dev/shm type tmpfs (rw,relatime,size=65536k)
/dev/sda1 on /var/config/tmux type ext4 (ro,relatime,commit=30)
/dev/sda1 on /etc/ssh/keys type ext4 (rw,relatime,commit=30)
/dev/sda1 on /var/lib/docker type ext4 (rw,relatime,commit=30)
/dev/sda1 on /run/google/devshell type ext4 (rw,relatime,commit=30)
none on /sys/kernel/security type securityfs (rw,relatime)
chandrusrinivasan11@cloudshell:~/msd$ sudo umount
umount: bad usage
Try 'umount --help' for more information.
chandrusrinivasan11@cloudshell:~/msd$ sudo blkid
/dev/sdb1: UUID="b9f0e455-8c03-483b-a0d6-629a47561fb9" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="5245a676-01"
/dev/dm-0: LABEL="ROOT-A" SEC_TYPE="ext2" BLOCK_SIZE="4096" TYPE="ext4"
/dev/sda4: PARTLABEL="KERN-B" PARTUUID="76f23ba4-cdd3-1143-91d2-9a5f7372c111"
/dev/sda2: PARTLABEL="KERN-A" PARTUUID="2762bea5-16e2-5944-b00a-f19370e6ab65"
/dev/sda12: SEC_TYPE="msdos" LABEL_FATBOOT="EFI-SYSTEM" LABEL="EFI-SYSTEM" UUID="7BE5-F6D0" BLOCK_SIZE="512" TYPE="vfat" PARTLABEL="EFI-SYSTEM" PARTUUID="1b9d2508-42"
/dev/sda9: PARTLABEL="reserved" PARTUUID="9634bb25-70d7-1a45-8b09-3648a7de5fe8"
/dev/sda10: PARTLABEL="reserved" PARTUUID="9f5dc8da-5cc5-f44e-8289-f69f69e82b1d"
/dev/sda7: PARTLABEL="ROOT-C" PARTUUID="9bf2b5b6-84ac-a94a-b087-360efbfff48b7"
/dev/sda5: PARTLABEL="ROOT-B" PARTUUID="969395aa-4a21-b54f-a5ae-3d373e0d8005"
/dev/sda3: LABEL="ROOT-A" SEC_TYPE="ext2" BLOCK_SIZE="4096" TYPE="ext4" PARTLABEL="ROOT-A" PARTUUID="cald3ffc-74bb-3a4c-872e-8923bf79698e"
/dev/sda1: LABEL="STATE" UUID="9f2dbcb8-9534-4465-bea2-481409599ab8" BLOCK_SIZE="4096" TYPE="ext4" PARTLABEL="STATE" PARTUUID="3154d2f0-8b5c-4447-a5e0-0c5002398a44"
/dev/sda11: PARTLABEL="RWF" PARTUUID="bfed8244-2507-ec48-adfc-b7aa9c3bbbae"
/dev/sda8: LABEL="OEM" UUID="7c475393-34ab-4c9b-aebb-a9cea72447d4" BLOCK_SIZE="4096" TYPE="ext4" PARTLABEL="OEM" PARTUUID="d2d9c06d-32cf-1b49-a5e8-e9c15ebb7dc5"
/dev/sda6: PARTLABEL="KERN-C" PARTUUID="9ea468af-861b-9946-a767-3bfe1912f69e"
chandrusrinivasan11@cloudshell:~/msd$
```

```
chandrusrinivasan11@cloudshell:~/msd$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0  100G  0 disk
├─sda1       8:1    0  95.8G  0 part /run/google/devshell
│            │      │      │      │      │ /var/lib/docker
│            │      │      │      │      │ /etc/ssh/keys
│            │      │      │      │      │ /var/config/tmux
│            │      │      │      │      │ /etc/resolv.conf
│            │      │      │      │      │ /etc/hostname
│            │      │      │      │      │ /etc/hosts
│            │      │      │      │      │ /root
├─sda2       8:2    0   16M  0 part
├─sda3       8:3    0    2G  0 part
├─sda4       8:4    0   16M  0 part
├─sda5       8:5    0    2G  0 part
├─sda6       8:6    0  512B  0 part
├─sda7       8:7    0  512B  0 part
├─sda8       8:8    0   16M  0 part
├─sda9       8:9    0  512B  0 part
├─sda10      8:10   0  512B  0 part
├─sda11      8:11   0    8M  0 part
├─sda12      8:12   0   32M  0 part
└─sdb        8:16   0    5G  0 disk
   └─sdb1     8:17   0    5G  0 part /home
```

```
chandrusrinivasan11@cloudshell:~/msd$ sudo tune2fs -l /dev/sda1
tune2fs 1.47.0 (5-Feb-2023)
```

```
Filesystem volume name:  STATE
Last mounted on:        /var
Filesystem UUID:         9f2dbcb8-9534-4465-bea2-481409599ab8
Filesystem magic number: 0xEF53
Filesystem revision #:   1 (dynamic)
Filesystem features:     has_journal ext_attr resize_inode dir_index filetype needs_recovery extent 64bit flex_bg
Filesystem flags:        signed_directory_hash
Default mount options:   user_xattr acl
Filesystem state:        clean
Errors behavior:         Remount read-only
Filesystem OS type:      Linux
Inode count:              6258720
Block count:              25126395
Reserved block count:    0
Overhead clusters:       418418
```

```
Lifetime writes:          77 GB
Reserved blocks uid:      0 (user root)
Reserved blocks gid:      0 (group root)
First inode:              11
Inode size:               256
Required extra isize:     32
Desired extra isize:      32
Journal inode:            8
Default directory hash:   half_md4
Directory Hash Seed:      239222de-2248-4ca5-8d76-167546212650
Journal backup:           inode blocks
Checksum type:            crc32c
Checksum:                 0xda887b0d
chandrusrinivasan11@cloudshell:~/msd$
```

```
top - 19:34:26 up 2:20, 0 user, load average: 0.03, 0.02, 0.00
Tasks: 38 total, 1 running, 37 sleeping, 0 stopped, 0 zombie
%Cpu(s): 2.2 us, 1.3 sy, 0.0 ni, 96.0 id, 0.2 wa, 0.0 hi, 0.2 si, 0.2 st
MiB Mem : 7947.4 total, 4139.0 free, 2462.2 used, 1590.5 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 5485.2 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
453	root	20	0	1793224	46472	32816	S	0.3	0.6	0:07.09	containerd
724	chandru+	10	-10	7988	3428	2004	S	0.3	0.0	0:00.23	tmux
3494	chandru+	20	0	42.6g	960660	62292	S	0.3	11.8	0:51.89	node
1	root	20	0	4324	3428	3148	S	0.0	0.0	0:00.07	bash
9	syslog	20	0	222140	3780	2420	S	0.0	0.0	0:00.07	rsyslogd
25	root	20	0	38768	28040	10932	S	0.0	0.3	0:01.47	python
26	root	20	0	6196	2572	2332	S	0.0	0.0	0:00.00	logger
67	root	10	-10	12024	4252	3140	S	0.0	0.1	0:00.09	sshd
318	root	20	0	2021836	84604	61424	S	0.0	1.0	0:01.12	dockerd
441	root	20	0	1234012	12392	8288	S	0.0	0.2	0:01.67	editor-proxy
442	root	20	0	16328	6352	5428	S	0.0	0.1	0:00.01	sudo
447	root	20	0	1226100	2696	1976	S	0.0	0.0	0:00.01	tmux-agent
646	root	20	0	2696	1596	1496	S	0.0	0.0	0:00.00	sleep
670	root	20	0	9196	4156	3780	S	0.0	0.1	0:00.00	runuser
673	chandru+	20	0	4324	3172	2908	S	0.0	0.0	0:00.00	sh
695	chandru+	20	0	11.3g	168440	51260	S	0.0	2.1	0:14.99	node
725	chandru+	10	-10	8800	5796	3924	S	0.0	0.1	0:00.21	bash
1599	chandru+	20	0	1350424	83100	48372	S	0.0	1.0	0:09.25	node
1610	chandru+	20	0	9020	6012	4000	S	0.0	0.1	0:00.07	bash
2083	chandru+	20	0	9020	6052	4036	S	0.0	0.1	0:00.05	bash
2549	chandru+	20	0	9020	6112	4096	S	0.0	0.1	0:00.07	bash
2551	chandru+	20	0	9020	6132	4116	S	0.0	0.1	0:00.07	bash
3452	chandru+	20	0	1048516	62388	47912	S	0.0	0.8	0:00.59	node
3479	root	10	-10	13996	9716	8340	S	0.0	0.1	0:00.02	sshd
3481	root	10	-10	13996	9612	8236	S	0.0	0.1	0:00.01	sshd
3483	chandru+	10	-10	14256	6452	4796	S	0.0	0.1	0:00.00	sshd
3484	chandru+	10	-10	14256	6372	4716	S	0.0	0.1	0:00.16	sshd
3485	chandru+	10	-10	7340	3796	3544	S	0.0	0.0	0:00.00	bash
3486	chandru+	10	-10	7340	3632	3380	S	0.0	0.0	0:00.00	bash
3489	chandru+	10	-10	7340	3572	3312	S	0.0	0.0	0:00.00	start-shell.sh
3490	chandru+	10	-10	7604	4288	3688	S	0.0	0.1	0:00.00	bash
3491	chandru+	10	-10	6880	3264	2968	S	0.0	0.0	0:00.00	tmux
3505	chandru+	20	0	1048516	61908	47804	S	0.0	0.8	0:00.60	node
3527	chandru+	20	0	1016392	62968	46420	S	0.0	0.8	0:00.52	node

0%

1%

Mem

Swp

2.0%

1.3%

2.17C/7.76C

OK/OK

Tasks: 36

116 thr

0 kthr

1 running

Load average: 0.02

0.02

0.00

Cpu

02:20:39

pid	user	pr	ni	virt	res	shd	cpu	mem	time	command
3933	chandrusr	10	0	1318K	21100	45172	0.7	0.1	0:00.04	/google/devshell/editor/code-oss-for-cloud-shell/ACde /google/devshell/editor/code-oss-for-cloud-shell/CUT/Bootstrap fork
1	root	20	0	1324	1428	1148	0.0	0.0	0:00.06	/bin/bash /google/scripts/onrun.sh sleep infinity
9	syslog	20	0	216K	3780	2420	0.0	0.0	0:00.00	/usr/sbin/rsyslogd
10	syslog	20	0	216K	3780	2420	0.0	0.0	0:00.04	/usr/sbin/rsyslogd
11	syslog	20	0	216K	3780	2420	0.0	0.0	0:00.00	/usr/sbin/rsyslogd
12	syslog	20	0	216K	3780	2420	0.0	0.0	0:00.03	/usr/sbin/rsyslogd
25	root	20	0	38768	28040	10932	0.0	0.3	0:01.47	/usr/bin/python /usr/bin/supervisord -n -c /google/devshell/supervisord.conf --pidfile=/var/run/supervisor.pid --logfile=/var
26	root	20	0	6196	2572	2332	0.0	0.0	0:00.00	logger -t supervisord
67	root	10	0	12024	4252	3140	0.0	0.1	0:00.09	sshd: /usr/sbin/sshd -p 22 -o AuthorizedKeysFile=/etc/ssh/keys/authorized keys [listener] 0 of 10-60 startups
318	root	20	0	1974K	84604	61424	0.0	1.0	0:00.17	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
369	root	20	0	1974K	84604	61424	0.0	1.0	0:00.17	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
370	root	20	0	1974K	84604	61424	0.0	1.0	0:00.00	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
372	root	20	0	1974K	84604	61424	0.0	1.0	0:00.15	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
375	root	20	0	1974K	84604	61424	0.0	1.0	0:00.00	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
441	root	20	0	1205K	12392	8288	0.0	0.2	0:00.45	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
442	root	20	0	16328	6352	5428	0.0	0.1	0:00.01	sudo /google/devshell/tmux-agent
443	root	20	0	1974K	84604	61424	0.0	1.0	0:00.00	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
444	root	20	0	1205K	12392	8288	0.0	0.2	0:00.34	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
445	root	20	0	1205K	12392	8288	0.0	0.2	0:00.53	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
446	root	20	0	1205K	12392	8288	0.0	0.2	0:00.29	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
447	root	20	0	1157K	2696	1976	0.0	0.0	0:00.00	/google/devshell/tmux-agent
448	root	20	0	1157K	2696	1976	0.0	0.0	0:00.00	/google/devshell/tmux-agent
449	root	20	0	1157K	2696	1976	0.0	0.0	0:00.00	/google/devshell/tmux-agent
450	root	20	0	1157K	2696	1976	0.0	0.0	0:00.00	/google/devshell/tmux-agent
451	root	20	0	1205K	12392	8288	0.0	0.2	0:00.02	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
453	root	20	0	1751K	46472	32816	0.0	0.6	0:00.04	containerd --config /var/run/docker/containerd/containerd.toml
454	root	20	0	1974K	84604	61424	0.0	1.0	0:00.00	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
455	root	20	0	1751K	46472	32816	0.0	0.6	0:02.01	containerd --config /var/run/docker/containerd/containerd.toml
456	root	20	0	1751K	46472	32816	0.0	0.6	0:00.00	containerd --config /var/run/docker/containerd/containerd.toml
457	root	20	0	1751K	46472	32816	0.0	0.6	0:00.31	containerd --config /var/run/docker/containerd/containerd.toml
458	root	20	0	1751K	46472	32816	0.0	0.6	0:00.00	containerd --config /var/run/docker/containerd/containerd.toml

F1

F2

F3

F4

F5

F6

F7

F8

F9

F10

```
3505 chandru+ 20 0 1048316 61908 47804 S 0.0 0.8 0:00.60 node
3527 chandru+ 20 0 1016392 62972 46420 S 0.0 0.8 0:00.52 node
```

```
chandrusrinivasan11@cloudshell:~/msd$ htop
chandrusrinivasan11@cloudshell:~/msd$ vmstat
```

```
procs -----memory----- ---swap-- -----io----- -system-- -----cpu-----
 r b  swpd  free  buff  cache  si  so  bi  bo  in  cs  us  sy  id  wa  st  gu
 1 0    0 4234804 82580 1547768  0  0  217  43 597   5  2  1 97  0  0  0
```

```
chandrusrinivasan11@cloudshell:~/msd$ free
```

	total	used	free	shared	buff/cache	available
Mem:	8138096	2523584	4234804	1148	1630404	5614512
Swap:	0	0	0			

```
chandrusrinivasan11@cloudshell:~/msd$ uptime
19:35:16 up 2:21, 0 user, load average: 0.01, 0.01, 0.00
chandrusrinivasan11@cloudshell:~/msd$ █
```

```
chandrusrinivasan11@cloudshell:~/msd$ sudo useradd chandru
chandrusrinivasan11@cloudshell:~/msd$ ls
play
chandrusrinivasan11@cloudshell:~/msd$ sudo userdel chandru
chandrusrinivasan11@cloudshell:~/msd$ ls
play
chandrusrinivasan11@cloudshell:~/msd$ groups
chandrusrinivasan11 adm sudo docker
chandrusrinivasan11@cloudshell:~/msd$ sudo groupadd developers
chandrusrinivasan11@cloudshell:~/msd$ ls
play
chandrusrinivasan11@cloudshell:~/msd$ sudo groupdel developers
chandrusrinivasan11@cloudshell:~/msd$ ls
play
chandrusrinivasan11@cloudshell:~/msd$
```



```
play
chandrusrinivasan11@cloudshell:~/msd$ bzip2 play

bzip2: I/O or other error, bailing out.  Possible reason follows.
bzip2: Operation not permitted
      Input file = play, output file = play.bz2
chandrusrinivasan11@cloudshell:~/msd$ bzip2 play.py
-bash: bzip2: command not found
chandrusrinivasan11@cloudshell:~/msd$ bzip2 play.py
bzip2: Can't open input file play.py: No such file or directory.
chandrusrinivasan11@cloudshell:~/msd$ bunzip2 file.txt.bz2
bunzip2: Can't open input file file.txt.bz2: No such file or directory.
chandrusrinivasan11@cloudshell:~/msd$ xz play
xz: play: Cannot remove: Operation not permitted
chandrusrinivasan11@cloudshell:~/msd$ tar -cvf archive.tar
tar: Cowardly refusing to create an empty archive
Try 'tar --help' or 'tar --usage' for more information.
chandrusrinivasan11@cloudshell:~/msd$ tar -xzvf archive.tar.gz
tar (child): archive.tar.gz: Cannot open: No such file or directory
tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
chandrusrinivasan11@cloudshell:~/msd$ cd file.txt
-bash: cd: file.txt: No such file or directory
chandrusrinivasan11@cloudshell:~/msd$ mkdir msd
chandrusrinivasan11@cloudshell:~/msd$ cd file.txt
-bash: cd: file.txt: No such file or directory
chandrusrinivasan11@cloudshell:~/msd$ zip archive.zip play
  adding: play (deflated 53%)
chandrusrinivasan11@cloudshell:~/msd$ unzip archive.zip
Archive:  archive.zip
replace play? [y]es, [n]o, [A]ll, [N]one, [r]ename: lzma play
error:  invalid response [lzma play]
replace play? [y]es, [n]o, [A]ll, [N]one, [r]ename: error:  invalid response [{ENTER}]
replace play? [y]es, [n]o, [A]ll, [N]one, [r]ename:
error:  invalid response [{ENTER}]
replace play? [y]es, [n]o, [A]ll, [N]one, [r]ename: ^Cchandrusrinivasan11@cloudshell:~/msd$
```